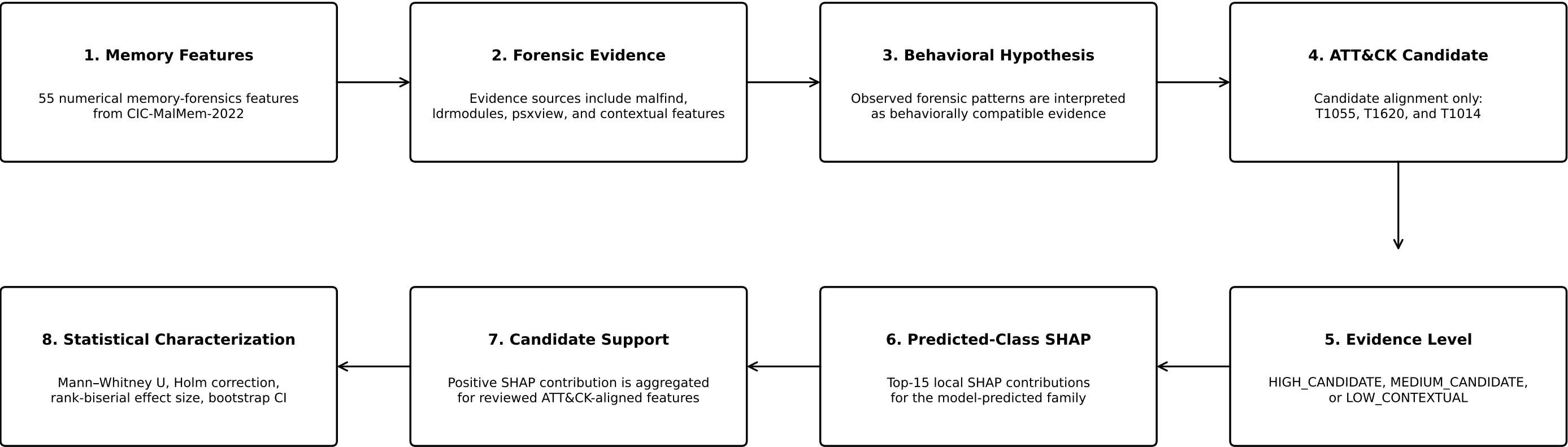


Evidence-Aware ATT&CK Alignment for Memory Malware Analysis

Forensic evidence is translated into behavioral hypotheses before ATT&CK-aligned attribution support is quantified.



$$S_k(x) = \frac{\sum_{i \in \mathcal{F}_k} \max(\phi_i^{(\hat{y})}(x), 0)}{\sum_{j \in \text{TopK}} \max(\phi_j^{(\hat{y})}(x), 0)}$$

$K = 15$; $\phi_i^{(\hat{y})}(x)$ is the local SHAP value for the predicted class; $\mathcal{F}_k$ is the set of reviewed features aligned with candidate k .

T1055 — Process Injection HIGH_CANDIDATE 2 mapped features Malware support: 41.0% Benign support: 0.0% r_{tb} =0.410	T1620 — Reflective Code Loading MEDIUM_CANDIDATE 6 mapped features Malware support: 93.9% Benign support: 45.6% r_{tb} =0.803	T1014 — Rootkit MEDIUM_CANDIDATE 14 mapped features Malware support: 37.9% Benign support: 7.2% r_{tb} =0.334
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Interpretation guardrail: ATT&CK alignment denotes evidence-supported behavioral hypotheses, not confirmed technique detection, causal attribution, or ground-truth ATT&CK labeling.